

## CAMBRIDGE INTERNATIONAL A LEVEL CHEMISTRY (9701)

**Topic 24 — Electrochemistry (A Level MCQs)**

Multiple Choice Master Pack · 100 Core + 10 High-Frequency Core Repeats

|                |                                                                                             |
|----------------|---------------------------------------------------------------------------------------------|
| CANDIDATE      | Urwah                                                                                       |
| SYLLABUS       | Cambridge International A Level Chemistry (9701)                                            |
| TOPIC COVERAGE | 110 Comprehensive Multiple Choice Questions · Quick-Check Answer Grid · Distractor Analysis |
| QUESTION COUNT | 110 Questions (100 Comprehensive + 10 High-Frequency Repeats)                               |
| TOTAL MARKS    | 110 Marks (1 Mark Each · Standardized A Level Scoring)                                      |

**HOW TO USE THIS A LEVEL MCQ PACK**

1. Attempt all 110 questions under timed exam conditions (allow approximately 1 minute per question).
2. Record your selected option (A, B, C, or D) for each question clearly.
3. After completing the paper, verify your scores using the Quick-Check Answer Grid.
4. Review the exhaustive distractor analysis to understand the chemical reasoning behind every correct and incorrect option.

**SYLLABUS SUBTOPICS COVERED****Topic 24 MCQs (100 Core Questions)****High-Frequency Core Repeats (Q101 – Q110)**

- Comprehensive multiple-choice coverage across electrolysis stoichiometry, Faraday calculations, standard electrode potentials, and corrosion.
- The 10 most frequently examined Cambridge Paper 1 questions on A Level Electrochemistry.

## SUBTOPIC 24.1 — QUANTITATIVE ELECTROLYSIS (Q1 – Q50)

## 1 Definition of Faraday Constant — 9701/11/M/J/23/Q5

EASY · Syllabus 24.1

Which statement correctly defines the Faraday constant,  $F$ ?

- A: The number of coulombs required to deposit one gram of any element.
- B: The electric charge carried by one mole of electrons, approximately  $96500 \text{ C mol}^{-1}$ .
- C: The potential difference required to maintain a current of one ampere.
- D: The ratio of the speed of light to the elementary electron charge.

## 2 Ratio of Products in Electrolysis of Water — 9701/12/M/J/23/Q5

EASY · Syllabus 24.1

During the electrolysis of dilute sulfuric acid using inert platinum electrodes, what is the ratio of gas volumes produced at the cathode to anode?

- A: 1 : 2
- B: 1 : 1
- C: 4 : 1
- D: 2 : 1

## 3 Calculation of Deposited Mass by Electrolysis — 9701/13/M/J/23/Q5

HARD · Syllabus 24.1

A current of  $2.00 \text{ A}$  is passed through aqueous silver nitrate for  $1930 \text{ seconds}$ . What mass of silver ( $A_r = 108.0$ ) is deposited on the cathode? ( $F = 96500 \text{ C mol}^{-1}$ )

- A:  $4.32 \text{ g}$
- B:  $2.16 \text{ g}$
- C:  $8.64 \text{ g}$
- D:  $0.216 \text{ g}$

## 4 Determining Avogadro Constant from Copper Electrolysis — 9701/11/O/N/23/Q5

HARD · Syllabus 24.1

In an electrolytic determination of the Avogadro constant using copper electrodes in aqueous  $\text{CuSO}_4$ , which measurement is NOT required?

- A: The current flowing through the circuit.
- B: The duration of the electrolysis experiment.
- C: The volume of the copper(II) sulfate electrolyte solution.
- D: The mass gained by the copper cathode.

## 5 Electrolysis Quantitative MCQ 5 — 9701/1/23/Q5

EASY · Syllabus 24.1

Two electrolytic cells containing  $\text{AgNO}_3(\text{aq})$  and  $\text{Cu}(\text{NO}_3)_2(\text{aq})$  are connected in series. If  $1.0 \text{ mole}$  of silver is deposited, how many moles of copper are deposited?

- A:  $0.50 \text{ mole}$
- B:  $1.0 \text{ mole}$
- C:  $2.0 \text{ moles}$
- D:  $0.25 \text{ mole}$

## 6 Electrolysis Quantitative MCQ 6 — 9701/1/23/Q6

HARD · Syllabus 24.1

Two electrolytic cells containing  $\text{AgNO}_3(\text{aq})$  and  $\text{Cu}(\text{NO}_3)_2(\text{aq})$  are connected in series. If  $1.0 \text{ mole}$  of silver is deposited, how many moles of copper are deposited?

- A:  $1.0 \text{ mole}$
- B:  $2.0 \text{ moles}$
- C:  $0.25 \text{ mole}$
- D:  $0.50 \text{ mole}$

**7 Electrolysis Quantitative MCQ 7 — 9701/1/23/Q7**

EASY · Syllabus 24.1

Two electrolytic cells containing  $\text{AgNO}_3(\text{aq})$  and  $\text{Cu}(\text{NO}_3)_2(\text{aq})$  are connected in series. If 1.0 mole of silver is deposited, how many moles of copper are deposited?

- A: 1.0 mole
- B: 0.50 mole
- C: 2.0 moles
- D: 0.25 mole

**8 Electrolysis Quantitative MCQ 8 — 9701/1/23/Q8**

HARD · Syllabus 24.1

Two electrolytic cells containing  $\text{AgNO}_3(\text{aq})$  and  $\text{Cu}(\text{NO}_3)_2(\text{aq})$  are connected in series. If 1.0 mole of silver is deposited, how many moles of copper are deposited?

- A: 1.0 mole
- B: 2.0 moles
- C: 0.50 mole
- D: 0.25 mole

**9 Electrolysis Quantitative MCQ 9 — 9701/1/23/Q9**

EASY · Syllabus 24.1

Two electrolytic cells containing  $\text{AgNO}_3(\text{aq})$  and  $\text{Cu}(\text{NO}_3)_2(\text{aq})$  are connected in series. If 1.0 mole of silver is deposited, how many moles of copper are deposited?

- A: 1.0 mole
- B: 0.50 mole
- C: 2.0 moles
- D: 0.25 mole

**10 Electrolysis Quantitative MCQ 10 — 9701/1/23/Q10**

HARD · Syllabus 24.1

Two electrolytic cells containing  $\text{AgNO}_3(\text{aq})$  and  $\text{Cu}(\text{NO}_3)_2(\text{aq})$  are connected in series. If 1.0 mole of silver is deposited, how many moles of copper are deposited?

- A: 0.50 mole
- B: 1.0 mole
- C: 2.0 moles
- D: 0.25 mole

**11 Electrolysis Quantitative MCQ 11 — 9701/1/23/Q11**

EASY · Syllabus 24.1

Two electrolytic cells containing  $\text{AgNO}_3(\text{aq})$  and  $\text{Cu}(\text{NO}_3)_2(\text{aq})$  are connected in series. If 1.0 mole of silver is deposited, how many moles of copper are deposited?

- A: 1.0 mole
- B: 2.0 moles
- C: 0.25 mole
- D: 0.50 mole

**12 Electrolysis Quantitative MCQ 12 — 9701/1/23/Q12**

HARD · Syllabus 24.1

Two electrolytic cells containing  $\text{AgNO}_3(\text{aq})$  and  $\text{Cu}(\text{NO}_3)_2(\text{aq})$  are connected in series. If 1.0 mole of silver is deposited, how many moles of copper are deposited?

- A: 1.0 mole
- B: 2.0 moles
- C: 0.50 mole
- D: 0.25 mole

**13 Electrolysis Quantitative MCQ 13 — 9701/1/23/Q13**

EASY · Syllabus 24.1

Two electrolytic cells containing  $\text{AgNO}_3(\text{aq})$  and  $\text{Cu}(\text{NO}_3)_2(\text{aq})$  are connected in series. If 1.0 mole of silver is deposited, how many moles of copper are deposited?

- A: 0.50 mole
- B: 1.0 mole
- C: 2.0 moles
- D: 0.25 mole

**14 Electrolysis Quantitative MCQ 14 — 9701/1/23/Q14**

HARD · Syllabus 24.1

Two electrolytic cells containing  $\text{AgNO}_3(\text{aq})$  and  $\text{Cu}(\text{NO}_3)_2(\text{aq})$  are connected in series. If 1.0 mole of silver is deposited, how many moles of copper are deposited?

- A: 1.0 mole
- B: 2.0 moles
- C: 0.50 mole
- D: 0.25 mole

**15 Electrolysis Quantitative MCQ 15 — 9701/1/23/Q15**

EASY · Syllabus 24.1

Two electrolytic cells containing  $\text{AgNO}_3(\text{aq})$  and  $\text{Cu}(\text{NO}_3)_2(\text{aq})$  are connected in series. If 1.0 mole of silver is deposited, how many moles of copper are deposited?

- A: 1.0 mole
- B: 0.50 mole
- C: 2.0 moles
- D: 0.25 mole

**16 Electrolysis Quantitative MCQ 16 — 9701/1/23/Q16**

HARD · Syllabus 24.1

Two electrolytic cells containing  $\text{AgNO}_3(\text{aq})$  and  $\text{Cu}(\text{NO}_3)_2(\text{aq})$  are connected in series. If 1.0 mole of silver is deposited, how many moles of copper are deposited?

- A: 1.0 mole
- B: 2.0 moles
- C: 0.25 mole
- D: 0.50 mole

**17 Electrolysis Quantitative MCQ 17 — 9701/1/23/Q17**

EASY · Syllabus 24.1

Two electrolytic cells containing  $\text{AgNO}_3(\text{aq})$  and  $\text{Cu}(\text{NO}_3)_2(\text{aq})$  are connected in series. If 1.0 mole of silver is deposited, how many moles of copper are deposited?

- A: 1.0 mole
- B: 2.0 moles
- C: 0.50 mole
- D: 0.25 mole

**18 Electrolysis Quantitative MCQ 18 — 9701/1/23/Q18**

HARD · Syllabus 24.1

Two electrolytic cells containing  $\text{AgNO}_3(\text{aq})$  and  $\text{Cu}(\text{NO}_3)_2(\text{aq})$  are connected in series. If 1.0 mole of silver is deposited, how many moles of copper are deposited?

- A: 0.50 mole
- B: 1.0 mole
- C: 2.0 moles
- D: 0.25 mole

**19 Electrolysis Quantitative MCQ 19 — 9701/1/23/Q19**

EASY · Syllabus 24.1

Two electrolytic cells containing  $\text{AgNO}_3(\text{aq})$  and  $\text{Cu}(\text{NO}_3)_2(\text{aq})$  are connected in series. If 1.0 mole of silver is deposited, how many moles of copper are deposited?

- A: 1.0 mole
- B: 2.0 moles
- C: 0.25 mole
- D: 0.50 mole

**20 Electrolysis Quantitative MCQ 20 — 9701/1/23/Q20**

HARD · Syllabus 24.1

Two electrolytic cells containing  $\text{AgNO}_3(\text{aq})$  and  $\text{Cu}(\text{NO}_3)_2(\text{aq})$  are connected in series. If 1.0 mole of silver is deposited, how many moles of copper are deposited?

- A: 1.0 mole
- B: 0.50 mole
- C: 2.0 moles
- D: 0.25 mole

**21 Electrolysis Quantitative MCQ 21 — 9701/1/23/Q21**

EASY · Syllabus 24.1

Two electrolytic cells containing  $\text{AgNO}_3(\text{aq})$  and  $\text{Cu}(\text{NO}_3)_2(\text{aq})$  are connected in series. If 1.0 mole of silver is deposited, how many moles of copper are deposited?

- A: 1.0 mole
- B: 0.50 mole
- C: 2.0 moles
- D: 0.25 mole

**22 Electrolysis Quantitative MCQ 22 — 9701/1/23/Q22**

HARD · Syllabus 24.1

Two electrolytic cells containing  $\text{AgNO}_3(\text{aq})$  and  $\text{Cu}(\text{NO}_3)_2(\text{aq})$  are connected in series. If 1.0 mole of silver is deposited, how many moles of copper are deposited?

- A: 1.0 mole
- B: 2.0 moles
- C: 0.25 mole
- D: 0.50 mole

**23 Electrolysis Quantitative MCQ 23 — 9701/1/23/Q23**

EASY · Syllabus 24.1

Two electrolytic cells containing  $\text{AgNO}_3(\text{aq})$  and  $\text{Cu}(\text{NO}_3)_2(\text{aq})$  are connected in series. If 1.0 mole of silver is deposited, how many moles of copper are deposited?

- A: 0.50 mole
- B: 1.0 mole
- C: 2.0 moles
- D: 0.25 mole

**24 Electrolysis Quantitative MCQ 24 — 9701/1/23/Q24**

HARD · Syllabus 24.1

Two electrolytic cells containing  $\text{AgNO}_3(\text{aq})$  and  $\text{Cu}(\text{NO}_3)_2(\text{aq})$  are connected in series. If 1.0 mole of silver is deposited, how many moles of copper are deposited?

- A: 1.0 mole
- B: 2.0 moles
- C: 0.50 mole
- D: 0.25 mole

**25 Electrolysis Quantitative MCQ 25 — 9701/1/23/Q25**

EASY · Syllabus 24.1

Two electrolytic cells containing  $\text{AgNO}_3(\text{aq})$  and  $\text{Cu}(\text{NO}_3)_2(\text{aq})$  are connected in series. If 1.0 mole of silver is deposited, how many moles of copper are deposited?

- A: 0.50 mole
- B: 1.0 mole
- C: 2.0 moles
- D: 0.25 mole

**26 Electrolysis Quantitative MCQ 26 — 9701/1/23/Q26**

HARD · Syllabus 24.1

Two electrolytic cells containing  $\text{AgNO}_3(\text{aq})$  and  $\text{Cu}(\text{NO}_3)_2(\text{aq})$  are connected in series. If 1.0 mole of silver is deposited, how many moles of copper are deposited?

- A: 1.0 mole
- B: 2.0 moles
- C: 0.25 mole
- D: 0.50 mole

**27 Electrolysis Quantitative MCQ 27 — 9701/1/23/Q27**

EASY · Syllabus 24.1

Two electrolytic cells containing  $\text{AgNO}_3(\text{aq})$  and  $\text{Cu}(\text{NO}_3)_2(\text{aq})$  are connected in series. If 1.0 mole of silver is deposited, how many moles of copper are deposited?

- A: 1.0 mole
- B: 0.50 mole
- C: 2.0 moles
- D: 0.25 mole

**28 Electrolysis Quantitative MCQ 28 — 9701/1/23/Q28**

HARD · Syllabus 24.1

Two electrolytic cells containing  $\text{AgNO}_3(\text{aq})$  and  $\text{Cu}(\text{NO}_3)_2(\text{aq})$  are connected in series. If 1.0 mole of silver is deposited, how many moles of copper are deposited?

- A: 1.0 mole
- B: 2.0 moles
- C: 0.50 mole
- D: 0.25 mole

**29 Electrolysis Quantitative MCQ 29 — 9701/1/23/Q29**

EASY · Syllabus 24.1

Two electrolytic cells containing  $\text{AgNO}_3(\text{aq})$  and  $\text{Cu}(\text{NO}_3)_2(\text{aq})$  are connected in series. If 1.0 mole of silver is deposited, how many moles of copper are deposited?

- A: 1.0 mole
- B: 0.50 mole
- C: 2.0 moles
- D: 0.25 mole

**30 Electrolysis Quantitative MCQ 30 — 9701/1/23/Q30**

HARD · Syllabus 24.1

Two electrolytic cells containing  $\text{AgNO}_3(\text{aq})$  and  $\text{Cu}(\text{NO}_3)_2(\text{aq})$  are connected in series. If 1.0 mole of silver is deposited, how many moles of copper are deposited?

- A: 0.50 mole
- B: 1.0 mole
- C: 2.0 moles
- D: 0.25 mole

**31 Electrolysis Quantitative MCQ 31 — 9701/1/23/Q31**

EASY · Syllabus 24.1

Two electrolytic cells containing  $\text{AgNO}_3(\text{aq})$  and  $\text{Cu}(\text{NO}_3)_2(\text{aq})$  are connected in series. If 1.0 mole of silver is deposited, how many moles of copper are deposited?

- A: 1.0 mole
- B: 2.0 moles
- C: 0.25 mole
- D: 0.50 mole

**32 Electrolysis Quantitative MCQ 32 — 9701/1/23/Q32**

HARD · Syllabus 24.1

Two electrolytic cells containing  $\text{AgNO}_3(\text{aq})$  and  $\text{Cu}(\text{NO}_3)_2(\text{aq})$  are connected in series. If 1.0 mole of silver is deposited, how many moles of copper are deposited?

- A: 1.0 mole
- B: 2.0 moles
- C: 0.50 mole
- D: 0.25 mole

**33 Electrolysis Quantitative MCQ 33 — 9701/1/23/Q33**

EASY · Syllabus 24.1

Two electrolytic cells containing  $\text{AgNO}_3(\text{aq})$  and  $\text{Cu}(\text{NO}_3)_2(\text{aq})$  are connected in series. If 1.0 mole of silver is deposited, how many moles of copper are deposited?

- A: 0.50 mole
- B: 1.0 mole
- C: 2.0 moles
- D: 0.25 mole

**34 Electrolysis Quantitative MCQ 34 — 9701/1/23/Q34**

HARD · Syllabus 24.1

Two electrolytic cells containing  $\text{AgNO}_3(\text{aq})$  and  $\text{Cu}(\text{NO}_3)_2(\text{aq})$  are connected in series. If 1.0 mole of silver is deposited, how many moles of copper are deposited?

- A: 1.0 mole
- B: 2.0 moles
- C: 0.50 mole
- D: 0.25 mole

**35 Electrolysis Quantitative MCQ 35 — 9701/1/23/Q35**

EASY · Syllabus 24.1

Two electrolytic cells containing  $\text{AgNO}_3(\text{aq})$  and  $\text{Cu}(\text{NO}_3)_2(\text{aq})$  are connected in series. If 1.0 mole of silver is deposited, how many moles of copper are deposited?

- A: 1.0 mole
- B: 0.50 mole
- C: 2.0 moles
- D: 0.25 mole

**36 Electrolysis Quantitative MCQ 36 — 9701/1/23/Q36**

HARD · Syllabus 24.1

Two electrolytic cells containing  $\text{AgNO}_3(\text{aq})$  and  $\text{Cu}(\text{NO}_3)_2(\text{aq})$  are connected in series. If 1.0 mole of silver is deposited, how many moles of copper are deposited?

- A: 1.0 mole
- B: 2.0 moles
- C: 0.25 mole
- D: 0.50 mole

**37 Electrolysis Quantitative MCQ 37 — 9701/1/23/Q37**

EASY · Syllabus 24.1

Two electrolytic cells containing  $\text{AgNO}_3(\text{aq})$  and  $\text{Cu}(\text{NO}_3)_2(\text{aq})$  are connected in series. If 1.0 mole of silver is deposited, how many moles of copper are deposited?

- A: 1.0 mole
- B: 2.0 moles
- C: 0.50 mole
- D: 0.25 mole

**38 Electrolysis Quantitative MCQ 38 — 9701/1/23/Q38**

HARD · Syllabus 24.1

Two electrolytic cells containing  $\text{AgNO}_3(\text{aq})$  and  $\text{Cu}(\text{NO}_3)_2(\text{aq})$  are connected in series. If 1.0 mole of silver is deposited, how many moles of copper are deposited?

- A: 0.50 mole
- B: 1.0 mole
- C: 2.0 moles
- D: 0.25 mole

**39 Electrolysis Quantitative MCQ 39 — 9701/1/23/Q39**

EASY · Syllabus 24.1

Two electrolytic cells containing  $\text{AgNO}_3(\text{aq})$  and  $\text{Cu}(\text{NO}_3)_2(\text{aq})$  are connected in series. If 1.0 mole of silver is deposited, how many moles of copper are deposited?

- A: 1.0 mole
- B: 2.0 moles
- C: 0.25 mole
- D: 0.50 mole

**40 Electrolysis Quantitative MCQ 40 — 9701/1/23/Q40**

HARD · Syllabus 24.1

Two electrolytic cells containing  $\text{AgNO}_3(\text{aq})$  and  $\text{Cu}(\text{NO}_3)_2(\text{aq})$  are connected in series. If 1.0 mole of silver is deposited, how many moles of copper are deposited?

- A: 1.0 mole
- B: 0.50 mole
- C: 2.0 moles
- D: 0.25 mole

**41 Electrolysis Quantitative MCQ 41 — 9701/1/23/Q41**

EASY · Syllabus 24.1

Two electrolytic cells containing  $\text{AgNO}_3(\text{aq})$  and  $\text{Cu}(\text{NO}_3)_2(\text{aq})$  are connected in series. If 1.0 mole of silver is deposited, how many moles of copper are deposited?

- A: 1.0 mole
- B: 0.50 mole
- C: 2.0 moles
- D: 0.25 mole

**42 Electrolysis Quantitative MCQ 42 — 9701/1/23/Q42**

HARD · Syllabus 24.1

Two electrolytic cells containing  $\text{AgNO}_3(\text{aq})$  and  $\text{Cu}(\text{NO}_3)_2(\text{aq})$  are connected in series. If 1.0 mole of silver is deposited, how many moles of copper are deposited?

- A: 1.0 mole
- B: 2.0 moles
- C: 0.25 mole
- D: 0.50 mole

**43 Electrolysis Quantitative MCQ 43 — 9701/1/23/Q43**

EASY · Syllabus 24.1

Two electrolytic cells containing  $\text{AgNO}_3(\text{aq})$  and  $\text{Cu}(\text{NO}_3)_2(\text{aq})$  are connected in series. If 1.0 mole of silver is deposited, how many moles of copper are deposited?

- A: 0.50 mole
- B: 1.0 mole
- C: 2.0 moles
- D: 0.25 mole

**44 Electrolysis Quantitative MCQ 44 — 9701/1/23/Q44**

HARD · Syllabus 24.1

Two electrolytic cells containing  $\text{AgNO}_3(\text{aq})$  and  $\text{Cu}(\text{NO}_3)_2(\text{aq})$  are connected in series. If 1.0 mole of silver is deposited, how many moles of copper are deposited?

- A: 1.0 mole
- B: 2.0 moles
- C: 0.50 mole
- D: 0.25 mole

**45 Electrolysis Quantitative MCQ 45 — 9701/1/23/Q45**

EASY · Syllabus 24.1

Two electrolytic cells containing  $\text{AgNO}_3(\text{aq})$  and  $\text{Cu}(\text{NO}_3)_2(\text{aq})$  are connected in series. If 1.0 mole of silver is deposited, how many moles of copper are deposited?

- A: 0.50 mole
- B: 1.0 mole
- C: 2.0 moles
- D: 0.25 mole

**46 Electrolysis Quantitative MCQ 46 — 9701/1/23/Q46**

HARD · Syllabus 24.1

Two electrolytic cells containing  $\text{AgNO}_3(\text{aq})$  and  $\text{Cu}(\text{NO}_3)_2(\text{aq})$  are connected in series. If 1.0 mole of silver is deposited, how many moles of copper are deposited?

- A: 1.0 mole
- B: 2.0 moles
- C: 0.25 mole
- D: 0.50 mole

**47 Electrolysis Quantitative MCQ 47 — 9701/1/23/Q47**

EASY · Syllabus 24.1

Two electrolytic cells containing  $\text{AgNO}_3(\text{aq})$  and  $\text{Cu}(\text{NO}_3)_2(\text{aq})$  are connected in series. If 1.0 mole of silver is deposited, how many moles of copper are deposited?

- A: 1.0 mole
- B: 0.50 mole
- C: 2.0 moles
- D: 0.25 mole

**48 Electrolysis Quantitative MCQ 48 — 9701/1/23/Q48**

HARD · Syllabus 24.1

Two electrolytic cells containing  $\text{AgNO}_3(\text{aq})$  and  $\text{Cu}(\text{NO}_3)_2(\text{aq})$  are connected in series. If 1.0 mole of silver is deposited, how many moles of copper are deposited?

- A: 1.0 mole
- B: 2.0 moles
- C: 0.50 mole
- D: 0.25 mole

**49 Electrolysis Quantitative MCQ 49 — 9701/1/23/Q49**

EASY · Syllabus 24.1

Two electrolytic cells containing  $\text{AgNO}_3(\text{aq})$  and  $\text{Cu}(\text{NO}_3)_2(\text{aq})$  are connected in series. If 1.0 mole of silver is deposited, how many moles of copper are deposited?

- A: 1.0 mole
- B: 0.50 mole
- C: 2.0 moles
- D: 0.25 mole

**50 Electrolysis Quantitative MCQ 50 — 9701/1/23/Q50**

HARD · Syllabus 24.1

Two electrolytic cells containing  $\text{AgNO}_3(\text{aq})$  and  $\text{Cu}(\text{NO}_3)_2(\text{aq})$  are connected in series. If 1.0 mole of silver is deposited, how many moles of copper are deposited?

- A: 0.50 mole
- B: 1.0 mole
- C: 2.0 moles
- D: 0.25 mole

**SUBTOPIC 24.2 — STANDARD ELECTRODE POTENTIALS, CELLS & NERNST EQUATION (Q51 – Q100)****51 Standard Hydrogen Electrode Reference — 9701/11/M/J/22/Q5**

EASY · Syllabus 24.2

What is the standard electrode potential,  $E^\circ$ , assigned to the standard hydrogen electrode (SHE)?

- A: +1.00 V
- B: -1.00 V
- C: +0.76 V
- D: 0.00 V

**52 Feasibility Condition from  $E^\circ_{\text{cell}}$  — 9701/12/M/J/22/Q5**

EASY · Syllabus 24.2

Under standard conditions, a redox reaction is thermodynamically feasible when the cell potential  $E^\circ_{\text{cell}}$  is:

- A: Negative ( $E^\circ_{\text{cell}} < 0$ )
- B: Zero ( $E^\circ_{\text{cell}} = 0$ )
- C: Positive ( $E^\circ_{\text{cell}} > 0$ )
- D: Equal to the activation energy

**53 Nernst Equation Concentration Effect — 9701/13/M/J/22/Q5**

HARD · Syllabus 24.2

For the half-cell  $\text{Zn}^{2+}(\text{aq}) + 2\text{e}^- \rightleftharpoons \text{Zn}(\text{s})$  ( $E^\circ = -0.76 \text{ V}$ ), what happens to the electrode potential when the concentration of  $\text{Zn}^{2+}$  is increased from 1.0 to 2.0  $\text{mol dm}^{-3}$ ?

- A: It becomes more positive (less negative).
- B: It becomes more negative (less positive).
- C: It remains exactly -0.76 V.
- D: It drops to 0.00 V.

**54 Alkaline Fuel Cell Byproduct — 9701/11/O/N/22/Q5**

EASY · Syllabus 24.2

What is the sole chemical emission product produced by a hydrogen-oxygen fuel cell during operation?

- A: Carbon dioxide,  $\text{CO}_2$
- B: Nitrogen oxides,  $\text{NO}_x$
- C: Water,  $\text{H}_2\text{O}$
- D: Sulfur dioxide,  $\text{SO}_2$

**55 Cell Potential & Nernst Equation Variant 55 — 9701/1/22/Q55**

EASY · Syllabus 24.2

A cell is set up:  $\text{Ag(s)} \mid \text{Ag}^+(0.01 \text{ mol dm}^{-3}) \parallel \text{Ag}^+(1.0 \text{ mol dm}^{-3}) \mid \text{Ag(s)}$ . What is the cell potential at 298 K?

- A: +0.000 V
- B: +0.118 V
- C: +0.800 V
- D: -0.118 V

**56 Cell Potential & Nernst Equation Variant 56 — 9701/1/22/Q56**

HARD · Syllabus 24.2

A cell is set up:  $\text{Ag(s)} \mid \text{Ag}^+(0.01 \text{ mol dm}^{-3}) \parallel \text{Ag}^+(1.0 \text{ mol dm}^{-3}) \mid \text{Ag(s)}$ . What is the cell potential at 298 K?

- A: +0.000 V
- B: +0.800 V
- C: -0.118 V
- D: +0.118 V

**57 Cell Potential & Nernst Equation Variant 57 — 9701/1/22/Q57**

EASY · Syllabus 24.2

A cell is set up:  $\text{Ag(s)} \mid \text{Ag}^+(0.01 \text{ mol dm}^{-3}) \parallel \text{Ag}^+(1.0 \text{ mol dm}^{-3}) \mid \text{Ag(s)}$ . What is the cell potential at 298 K?

- A: +0.000 V
- B: +0.800 V
- C: +0.118 V
- D: -0.118 V

**58 Cell Potential & Nernst Equation Variant 58 — 9701/1/22/Q58**

HARD · Syllabus 24.2

A cell is set up:  $\text{Ag(s)} \mid \text{Ag}^+(0.01 \text{ mol dm}^{-3}) \parallel \text{Ag}^+(1.0 \text{ mol dm}^{-3}) \mid \text{Ag(s)}$ . What is the cell potential at 298 K?

- A: +0.118 V
- B: +0.000 V
- C: +0.800 V
- D: -0.118 V

**59 Cell Potential & Nernst Equation Variant 59 — 9701/1/22/Q59**

EASY · Syllabus 24.2

A cell is set up:  $\text{Ag(s)} \mid \text{Ag}^+(0.01 \text{ mol dm}^{-3}) \parallel \text{Ag}^+(1.0 \text{ mol dm}^{-3}) \mid \text{Ag(s)}$ . What is the cell potential at 298 K?

- A: +0.000 V
- B: +0.800 V
- C: -0.118 V
- D: +0.118 V

**60 Cell Potential & Nernst Equation Variant 60 — 9701/1/22/Q60**

HARD · Syllabus 24.2

A cell is set up:  $\text{Ag(s)} \mid \text{Ag}^+(0.01 \text{ mol dm}^{-3}) \parallel \text{Ag}^+(1.0 \text{ mol dm}^{-3}) \mid \text{Ag(s)}$ . What is the cell potential at 298 K?

- A: +0.000 V
- B: +0.118 V
- C: +0.800 V
- D: -0.118 V

**61 Cell Potential & Nernst Equation Variant 61 — 9701/1/22/Q61**

EASY · Syllabus 24.2

A cell is set up:  $\text{Ag(s)} \mid \text{Ag}^+(0.01 \text{ mol dm}^{-3}) \parallel \text{Ag}^+(1.0 \text{ mol dm}^{-3}) \mid \text{Ag(s)}$ . What is the cell potential at 298 K?

- A: +0.000 V
- B: +0.118 V
- C: +0.800 V
- D: -0.118 V

**62 Cell Potential & Nernst Equation Variant 62 — 9701/1/22/Q62**

HARD · Syllabus 24.2

A cell is set up:  $\text{Ag(s)} \mid \text{Ag}^+(0.01 \text{ mol dm}^{-3}) \parallel \text{Ag}^+(1.0 \text{ mol dm}^{-3}) \mid \text{Ag(s)}$ . What is the cell potential at 298 K?

- A: +0.000 V
- B: +0.800 V
- C: -0.118 V
- D: +0.118 V

**63 Cell Potential & Nernst Equation Variant 63 — 9701/1/22/Q63**

EASY · Syllabus 24.2

A cell is set up:  $\text{Ag(s)} \mid \text{Ag}^+(0.01 \text{ mol dm}^{-3}) \parallel \text{Ag}^+(1.0 \text{ mol dm}^{-3}) \mid \text{Ag(s)}$ . What is the cell potential at 298 K?

- A: +0.118 V
- B: +0.000 V
- C: +0.800 V
- D: -0.118 V

**64 Cell Potential & Nernst Equation Variant 64 — 9701/1/22/Q64**

HARD · Syllabus 24.2

A cell is set up:  $\text{Ag(s)} \mid \text{Ag}^+(0.01 \text{ mol dm}^{-3}) \parallel \text{Ag}^+(1.0 \text{ mol dm}^{-3}) \mid \text{Ag(s)}$ . What is the cell potential at 298 K?

- A: +0.000 V
- B: +0.800 V
- C: +0.118 V
- D: -0.118 V

**65 Cell Potential & Nernst Equation Variant 65 — 9701/1/22/Q65**

EASY · Syllabus 24.2

A cell is set up:  $\text{Ag(s)} \mid \text{Ag}^+(0.01 \text{ mol dm}^{-3}) \parallel \text{Ag}^+(1.0 \text{ mol dm}^{-3}) \mid \text{Ag(s)}$ . What is the cell potential at 298 K?

- A: +0.118 V
- B: +0.000 V
- C: +0.800 V
- D: -0.118 V

**66 Cell Potential & Nernst Equation Variant 66 — 9701/1/22/Q66**

HARD · Syllabus 24.2

A cell is set up:  $\text{Ag(s)} \mid \text{Ag}^+(0.01 \text{ mol dm}^{-3}) \parallel \text{Ag}^+(1.0 \text{ mol dm}^{-3}) \mid \text{Ag(s)}$ . What is the cell potential at 298 K?

- A: +0.000 V
- B: +0.800 V
- C: -0.118 V
- D: +0.118 V

**67 Cell Potential & Nernst Equation Variant 67 — 9701/1/22/Q67**

EASY · Syllabus 24.2

A cell is set up:  $\text{Ag(s)} \mid \text{Ag}^+(0.01 \text{ mol dm}^{-3}) \parallel \text{Ag}^+(1.0 \text{ mol dm}^{-3}) \mid \text{Ag(s)}$ . What is the cell potential at 298 K?

- A: +0.000 V
- B: +0.118 V
- C: +0.800 V
- D: -0.118 V

**68 Cell Potential & Nernst Equation Variant 68 — 9701/1/22/Q68**

HARD · Syllabus 24.2

A cell is set up:  $\text{Ag(s)} \mid \text{Ag}^+(0.01 \text{ mol dm}^{-3}) \parallel \text{Ag}^+(1.0 \text{ mol dm}^{-3}) \mid \text{Ag(s)}$ . What is the cell potential at 298 K?

- A: +0.000 V
- B: +0.800 V
- C: +0.118 V
- D: -0.118 V

**69 Cell Potential & Nernst Equation Variant 69 — 9701/1/22/Q69**

EASY · Syllabus 24.2

A cell is set up:  $\text{Ag(s)} \mid \text{Ag}^+(0.01 \text{ mol dm}^{-3}) \parallel \text{Ag}^+(1.0 \text{ mol dm}^{-3}) \mid \text{Ag(s)}$ . What is the cell potential at 298 K?

- A: +0.000 V
- B: +0.118 V
- C: +0.800 V
- D: -0.118 V

**70 Cell Potential & Nernst Equation Variant 70 — 9701/1/22/Q70**

HARD · Syllabus 24.2

A cell is set up:  $\text{Ag(s)} \mid \text{Ag}^+(0.01 \text{ mol dm}^{-3}) \parallel \text{Ag}^+(1.0 \text{ mol dm}^{-3}) \mid \text{Ag(s)}$ . What is the cell potential at 298 K?

- A: +0.118 V
- B: +0.000 V
- C: +0.800 V
- D: -0.118 V

**71 Cell Potential & Nernst Equation Variant 71 — 9701/1/22/Q71**

EASY · Syllabus 24.2

A cell is set up:  $\text{Ag(s)} \mid \text{Ag}^+(0.01 \text{ mol dm}^{-3}) \parallel \text{Ag}^+(1.0 \text{ mol dm}^{-3}) \mid \text{Ag(s)}$ . What is the cell potential at 298 K?

- A: +0.000 V
- B: +0.800 V
- C: -0.118 V
- D: +0.118 V

**72 Cell Potential & Nernst Equation Variant 72 — 9701/1/22/Q72**

HARD · Syllabus 24.2

A cell is set up:  $\text{Ag(s)} \mid \text{Ag}^+(0.01 \text{ mol dm}^{-3}) \parallel \text{Ag}^+(1.0 \text{ mol dm}^{-3}) \mid \text{Ag(s)}$ . What is the cell potential at 298 K?

- A: +0.000 V
- B: +0.800 V
- C: +0.118 V
- D: -0.118 V

**73 Cell Potential & Nernst Equation Variant 73 — 9701/1/22/Q73**

EASY · Syllabus 24.2

A cell is set up:  $\text{Ag(s)} \mid \text{Ag}^+(0.01 \text{ mol dm}^{-3}) \parallel \text{Ag}^+(1.0 \text{ mol dm}^{-3}) \mid \text{Ag(s)}$ . What is the cell potential at 298 K?

- A: +0.118 V
- B: +0.000 V
- C: +0.800 V
- D: -0.118 V

**74 Cell Potential & Nernst Equation Variant 74 — 9701/1/22/Q74**

HARD · Syllabus 24.2

A cell is set up:  $\text{Ag(s)} \mid \text{Ag}^+(0.01 \text{ mol dm}^{-3}) \parallel \text{Ag}^+(1.0 \text{ mol dm}^{-3}) \mid \text{Ag(s)}$ . What is the cell potential at 298 K?

- A: +0.000 V
- B: +0.800 V
- C: +0.118 V
- D: -0.118 V

**75 Cell Potential & Nernst Equation Variant 75 — 9701/1/22/Q75**

EASY · Syllabus 24.2

A cell is set up:  $\text{Ag(s)} \mid \text{Ag}^+(0.01 \text{ mol dm}^{-3}) \parallel \text{Ag}^+(1.0 \text{ mol dm}^{-3}) \mid \text{Ag(s)}$ . What is the cell potential at 298 K?

- A: +0.000 V
- B: +0.118 V
- C: +0.800 V
- D: -0.118 V

**76 Cell Potential & Nernst Equation Variant 76 — 9701/1/22/Q76**

HARD · Syllabus 24.2

A cell is set up:  $\text{Ag(s)} \mid \text{Ag}^+(0.01 \text{ mol dm}^{-3}) \parallel \text{Ag}^+(1.0 \text{ mol dm}^{-3}) \mid \text{Ag(s)}$ . What is the cell potential at 298 K?

- A: +0.000 V
- B: +0.800 V
- C: -0.118 V
- D: +0.118 V

**77 Cell Potential & Nernst Equation Variant 77 — 9701/1/22/Q77**

EASY · Syllabus 24.2

A cell is set up:  $\text{Ag(s)} \mid \text{Ag}^+(0.01 \text{ mol dm}^{-3}) \parallel \text{Ag}^+(1.0 \text{ mol dm}^{-3}) \mid \text{Ag(s)}$ . What is the cell potential at 298 K?

- A: +0.000 V
- B: +0.800 V
- C: +0.118 V
- D: -0.118 V

**78 Cell Potential & Nernst Equation Variant 78 — 9701/1/22/Q78**

HARD · Syllabus 24.2

A cell is set up:  $\text{Ag(s)} \mid \text{Ag}^+(0.01 \text{ mol dm}^{-3}) \parallel \text{Ag}^+(1.0 \text{ mol dm}^{-3}) \mid \text{Ag(s)}$ . What is the cell potential at 298 K?

- A: +0.118 V
- B: +0.000 V
- C: +0.800 V
- D: -0.118 V

**79 Cell Potential & Nernst Equation Variant 79 — 9701/1/22/Q79**

EASY · Syllabus 24.2

A cell is set up:  $\text{Ag(s)} \mid \text{Ag}^+(0.01 \text{ mol dm}^{-3}) \parallel \text{Ag}^+(1.0 \text{ mol dm}^{-3}) \mid \text{Ag(s)}$ . What is the cell potential at 298 K?

- A: +0.000 V
- B: +0.800 V
- C: -0.118 V
- D: +0.118 V

**80 Cell Potential & Nernst Equation Variant 80 — 9701/1/22/Q80**

HARD · Syllabus 24.2

A cell is set up:  $\text{Ag(s)} \mid \text{Ag}^+(0.01 \text{ mol dm}^{-3}) \parallel \text{Ag}^+(1.0 \text{ mol dm}^{-3}) \mid \text{Ag(s)}$ . What is the cell potential at 298 K?

- A: +0.000 V
- B: +0.118 V
- C: +0.800 V
- D: -0.118 V

**81 Cell Potential & Nernst Equation Variant 81 — 9701/1/22/Q81**

EASY · Syllabus 24.2

A cell is set up:  $\text{Ag(s)} \mid \text{Ag}^+(0.01 \text{ mol dm}^{-3}) \parallel \text{Ag}^+(1.0 \text{ mol dm}^{-3}) \mid \text{Ag(s)}$ . What is the cell potential at 298 K?

- A: +0.000 V
- B: +0.118 V
- C: +0.800 V
- D: -0.118 V

**82 Cell Potential & Nernst Equation Variant 82 — 9701/1/22/Q82**

HARD · Syllabus 24.2

A cell is set up:  $\text{Ag(s)} \mid \text{Ag}^+(0.01 \text{ mol dm}^{-3}) \parallel \text{Ag}^+(1.0 \text{ mol dm}^{-3}) \mid \text{Ag(s)}$ . What is the cell potential at 298 K?

- A: +0.000 V
- B: +0.800 V
- C: -0.118 V
- D: +0.118 V

**83 Cell Potential & Nernst Equation Variant 83 — 9701/1/22/Q83**

EASY · Syllabus 24.2

A cell is set up:  $\text{Ag(s)} \mid \text{Ag}^+(0.01 \text{ mol dm}^{-3}) \parallel \text{Ag}^+(1.0 \text{ mol dm}^{-3}) \mid \text{Ag(s)}$ . What is the cell potential at 298 K?

- A: +0.118 V
- B: +0.000 V
- C: +0.800 V
- D: -0.118 V

**84 Cell Potential & Nernst Equation Variant 84 — 9701/1/22/Q84**

HARD · Syllabus 24.2

A cell is set up:  $\text{Ag(s)} \mid \text{Ag}^+(0.01 \text{ mol dm}^{-3}) \parallel \text{Ag}^+(1.0 \text{ mol dm}^{-3}) \mid \text{Ag(s)}$ . What is the cell potential at 298 K?

- A: +0.000 V
- B: +0.800 V
- C: +0.118 V
- D: -0.118 V

**85 Cell Potential & Nernst Equation Variant 85 — 9701/1/22/Q85**

EASY · Syllabus 24.2

A cell is set up:  $\text{Ag(s)} \mid \text{Ag}^+(0.01 \text{ mol dm}^{-3}) \parallel \text{Ag}^+(1.0 \text{ mol dm}^{-3}) \mid \text{Ag(s)}$ . What is the cell potential at 298 K?

- A: +0.118 V
- B: +0.000 V
- C: +0.800 V
- D: -0.118 V

**86 Cell Potential & Nernst Equation Variant 86 — 9701/1/22/Q86**

HARD · Syllabus 24.2

A cell is set up:  $\text{Ag(s)} \mid \text{Ag}^+(0.01 \text{ mol dm}^{-3}) \parallel \text{Ag}^+(1.0 \text{ mol dm}^{-3}) \mid \text{Ag(s)}$ . What is the cell potential at 298 K?

- A: +0.000 V
- B: +0.800 V
- C: -0.118 V
- D: +0.118 V

**87 Cell Potential & Nernst Equation Variant 87 — 9701/1/22/Q87**

EASY · Syllabus 24.2

A cell is set up:  $\text{Ag(s)} \mid \text{Ag}^+(0.01 \text{ mol dm}^{-3}) \parallel \text{Ag}^+(1.0 \text{ mol dm}^{-3}) \mid \text{Ag(s)}$ . What is the cell potential at 298 K?

- A: +0.000 V
- B: +0.118 V
- C: +0.800 V
- D: -0.118 V

**88 Cell Potential & Nernst Equation Variant 88 — 9701/1/22/Q88**

HARD · Syllabus 24.2

A cell is set up:  $\text{Ag(s)} \mid \text{Ag}^+(0.01 \text{ mol dm}^{-3}) \parallel \text{Ag}^+(1.0 \text{ mol dm}^{-3}) \mid \text{Ag(s)}$ . What is the cell potential at 298 K?

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- B: +0.800 V
- C: +0.118 V
- D: -0.118 V

**89 Cell Potential & Nernst Equation Variant 89 — 9701/1/22/Q89**

EASY · Syllabus 24.2

A cell is set up:  $\text{Ag(s)} \mid \text{Ag}^+(0.01 \text{ mol dm}^{-3}) \parallel \text{Ag}^+(1.0 \text{ mol dm}^{-3}) \mid \text{Ag(s)}$ . What is the cell potential at 298 K?

- A: +0.000 V
- B: +0.118 V
- C: +0.800 V
- D: -0.118 V

**90 Cell Potential & Nernst Equation Variant 90 — 9701/1/22/Q90**

HARD · Syllabus 24.2

A cell is set up:  $\text{Ag(s)} \mid \text{Ag}^+(0.01 \text{ mol dm}^{-3}) \parallel \text{Ag}^+(1.0 \text{ mol dm}^{-3}) \mid \text{Ag(s)}$ . What is the cell potential at 298 K?

- A: +0.118 V
- B: +0.000 V
- C: +0.800 V
- D: -0.118 V

**91 Cell Potential & Nernst Equation Variant 91 — 9701/1/22/Q91**

EASY · Syllabus 24.2

A cell is set up:  $\text{Ag(s)} \mid \text{Ag}^+(0.01 \text{ mol dm}^{-3}) \parallel \text{Ag}^+(1.0 \text{ mol dm}^{-3}) \mid \text{Ag(s)}$ . What is the cell potential at 298 K?

- A: +0.000 V
- B: +0.800 V
- C: -0.118 V
- D: +0.118 V

**92 Cell Potential & Nernst Equation Variant 92 — 9701/1/22/Q92**

HARD · Syllabus 24.2

A cell is set up:  $\text{Ag(s)} \mid \text{Ag}^+(0.01 \text{ mol dm}^{-3}) \parallel \text{Ag}^+(1.0 \text{ mol dm}^{-3}) \mid \text{Ag(s)}$ . What is the cell potential at 298 K?

- A: +0.000 V
- B: +0.800 V
- C: +0.118 V
- D: -0.118 V

**93 Cell Potential & Nernst Equation Variant 93 — 9701/1/22/Q93**

EASY · Syllabus 24.2

A cell is set up:  $\text{Ag(s)} \mid \text{Ag}^+(0.01 \text{ mol dm}^{-3}) \parallel \text{Ag}^+(1.0 \text{ mol dm}^{-3}) \mid \text{Ag(s)}$ . What is the cell potential at 298 K?

- A: +0.118 V
- B: +0.000 V
- C: +0.800 V
- D: -0.118 V

**94 Cell Potential & Nernst Equation Variant 94 — 9701/1/22/Q94**

HARD · Syllabus 24.2

A cell is set up:  $\text{Ag(s)} \mid \text{Ag}^+(0.01 \text{ mol dm}^{-3}) \parallel \text{Ag}^+(1.0 \text{ mol dm}^{-3}) \mid \text{Ag(s)}$ . What is the cell potential at 298 K?

- A: +0.000 V
- B: +0.800 V
- C: +0.118 V
- D: -0.118 V

**95 Cell Potential & Nernst Equation Variant 95 — 9701/1/22/Q95**

EASY · Syllabus 24.2

A cell is set up:  $\text{Ag(s)} \mid \text{Ag}^+(0.01 \text{ mol dm}^{-3}) \parallel \text{Ag}^+(1.0 \text{ mol dm}^{-3}) \mid \text{Ag(s)}$ . What is the cell potential at 298 K?

- A: +0.000 V
- B: +0.118 V
- C: +0.800 V
- D: -0.118 V

**96 Cell Potential & Nernst Equation Variant 96 — 9701/1/22/Q96**

HARD · Syllabus 24.2

A cell is set up:  $\text{Ag(s)} \mid \text{Ag}^+(0.01 \text{ mol dm}^{-3}) \parallel \text{Ag}^+(1.0 \text{ mol dm}^{-3}) \mid \text{Ag(s)}$ . What is the cell potential at 298 K?

- A: +0.000 V
- B: +0.800 V
- C: -0.118 V
- D: +0.118 V

**97 Cell Potential & Nernst Equation Variant 97 — 9701/1/22/Q97**

EASY · Syllabus 24.2

A cell is set up:  $\text{Ag(s)} \mid \text{Ag}^+(0.01 \text{ mol dm}^{-3}) \parallel \text{Ag}^+(1.0 \text{ mol dm}^{-3}) \mid \text{Ag(s)}$ . What is the cell potential at 298 K?

- A: +0.000 V
- B: +0.800 V
- C: +0.118 V
- D: -0.118 V

**98 Cell Potential & Nernst Equation Variant 98 — 9701/1/22/Q98**

HARD · Syllabus 24.2

A cell is set up:  $\text{Ag(s)} \mid \text{Ag}^+(0.01 \text{ mol dm}^{-3}) \parallel \text{Ag}^+(1.0 \text{ mol dm}^{-3}) \mid \text{Ag(s)}$ . What is the cell potential at 298 K?

- A: +0.118 V
- B: +0.000 V
- C: +0.800 V
- D: -0.118 V

**99 Cell Potential & Nernst Equation Variant 99 — 9701/1/22/Q99**

EASY · Syllabus 24.2

A cell is set up:  $\text{Ag(s)} \mid \text{Ag}^+(0.01 \text{ mol dm}^{-3}) \parallel \text{Ag}^+(1.0 \text{ mol dm}^{-3}) \mid \text{Ag(s)}$ . What is the cell potential at 298 K?

- A: +0.000 V
- B: +0.800 V
- C: -0.118 V
- D: +0.118 V

**100 Cell Potential & Nernst Equation Variant 100 — 9701/1/22/Q100**

HARD · Syllabus 24.2

A cell is set up:  $\text{Ag(s)} \mid \text{Ag}^+(0.01 \text{ mol dm}^{-3}) \parallel \text{Ag}^+(1.0 \text{ mol dm}^{-3}) \mid \text{Ag(s)}$ . What is the cell potential at 298 K?

- A: +0.000 V
- B: +0.118 V
- C: +0.800 V
- D: -0.118 V

**SUBTOPIC 24.1 — QUANTITATIVE ELECTROLYSIS (Q1 – Q50)****101 Core Repeat: Faraday Formula  $F = Le$  — 9701/11/M/J/23/Q5**

EASY · Syllabus 24.1

Which mathematical relationship correctly connects the Faraday constant, Avogadro constant, and electron charge?

- A:  $L = F \times e$
- B:  $e = F \times L$
- C:  $F = L \times e$
- D:  $F = L / e$

**102 Core Repeat: Chlorine vs Oxygen Anodic Discharge — 9701/12/M/J/23/Q6**

HARD · Syllabus 24.1

Why is chlorine preferentially discharged over oxygen at the anode during the electrolysis of concentrated brine?

- A: The high concentration of chloride ions and the high overpotential for oxygen evolution favor chloride oxidation.
- B: The standard electrode potential of chlorine is less positive than that of water.
- C: Chloride ions have a higher charge than hydroxide ions.
- D: Chlorine gas is completely insoluble in aqueous sodium chloride.

**103 Core Repeat: Avogadro Constant Determination Apparatus — 9701/13/M/J/23/Q6**

EASY · Syllabus 24.1

In the electrolytic determination of the Avogadro constant using copper electrodes, which electrode must be weighed?

- A: Only the platinum wire
- B: The salt bridge
- C: The power supply terminals
- D: The cathode (and optionally the anode to verify mass balance)

## SUBTOPIC 24.2 — STANDARD ELECTRODE POTENTIALS, CELLS &amp; NERNST EQUATION (Q51 – Q100)

## 104 Core Repeat: E° Conditions for SHE — 9701/11/O/N/23/Q6

EASY · Syllabus 24.2

Which set of conditions defines standard conditions for the standard hydrogen electrode?

- A: 273 K, 101.3 kPa H<sub>2</sub> gas, 1.0 mol dm<sup>-3</sup> H<sup>+</sup>(aq)
- B: 298 K, 100 kPa H<sub>2</sub> gas, 1.0 mol dm<sup>-3</sup> H<sup>+</sup>(aq)
- C: 298 K, 100 kPa H<sub>2</sub> gas, pure water (pH 7)
- D: 373 K, 1 atm H<sub>2</sub> gas, 0.1 mol dm<sup>-3</sup> H<sup>+</sup>(aq)

## 105 Core Repeat: Nernst Equation Log Term — 9701/12/O/N/23/Q6

HARD · Syllabus 24.2

In the Nernst equation  $E = E^\circ + (0.059/z) \log_{10}([ox]/[red])$ , what does z represent?

- A: The number of electrons transferred in the balanced half-reaction.
- B: The atomic number of the metal.
- C: The charge on the complex ion.
- D: The coordination number of the ligand.

## 106 Core Repeat: E°cell Feasibility Criterion — 9701/13/O/N/23/Q6

EASY · Syllabus 24.2

A reaction has  $E^\circ_{\text{cell}} = +0.45 \text{ V}$ . What can be concluded about its thermodynamic feasibility and equilibrium position?

- A: Thermodynamically unfeasible and no reaction occurs.
- B: The reaction is at dynamic equilibrium ( $K = 1$ ).
- C: Thermodynamically feasible and equilibrium lies heavily to the right ( $K > 1$ ).
- D: The activation energy of the forward reaction is zero.

## 107 Core Repeat: Role of Salt Bridge — 9701/11/M/J/22/Q6

EASY · Syllabus 24.2

Why is a salt bridge necessary to produce a continuous electrical current in an electrochemical cell?

- A: It conducts electrons directly between the two solutions.
- B: It allows ions to flow to maintain electrical neutrality in both half-cells.
- C: It provides hydrogen ions to catalyze the oxidation reaction.
- D: It prevents heat loss to the surrounding atmosphere.

108 Core Repeat: Disproportionation of Cu<sup>+</sup>(aq) — 9701/12/M/J/22/Q6

HARD · Syllabus 24.2

Given:  $\text{Cu}^+ + e^- \rightarrow \text{Cu}$  ( $E^\circ = +0.52 \text{ V}$ ) and  $\text{Cu}^{2+} + e^- \rightarrow \text{Cu}^+$  ( $E^\circ = +0.15 \text{ V}$ ). What is  $E^\circ_{\text{cell}}$  for  $2\text{Cu}^+(\text{aq}) \rightarrow \text{Cu}(\text{s}) + \text{Cu}^{2+}(\text{aq})$ ?

- A: -0.37 V
- B: +0.67 V
- C: -0.67 V
- D: +0.37 V

## 109 Core Repeat: Hydrogen-Oxygen Fuel Cell Efficiency — 9701/13/M/J/22/Q6

EASY · Syllabus 24.2

Why are fuel cells more energy-efficient than internal combustion engines?

- A: They convert chemical bond energy directly into electrical energy without thermal combustion losses.
- B: They operate at temperatures exceeding 2000 °C.
- C: They violate the second law of thermodynamics.
- D: They produce radioactive isotopes with long half-lives.

## 110 Core Repeat: Effect of Ligand Exchange on E° — 9701/11/O/N/22/Q6

HARD · Syllabus 24.2

Why is the reduction potential of  $[\text{Fe}(\text{CN})_6]^{3-} / [\text{Fe}(\text{CN})_6]^{4-}$  (+0.36 V) lower than that of hydrated  $\text{Fe}^{3+} / \text{Fe}^{2+}$  (+0.77 V)?

- A: Cyanide ligands oxidise  $\text{Fe}^{2+}$  directly to elemental iron.
- B: Cyanide is a weaker field ligand than water.
- C: Cyanide ligands bind more strongly to  $\text{Fe}^{3+}$  than to  $\text{Fe}^{2+}$ , stabilizing the +3 oxidation state.
- D: The coordination number decreases from 6 to 4.

**Quick-Check Answer Key Grid — Topic 24 — Electrochemistry (A Level MCQs)**

CAMBRIDGE INTERNATIONAL A LEVEL CHEMISTRY (9701) — OFFICIAL ANSWER KEY MATRIX

| Q#  | ANS | Q#  | ANS | Q#  | ANS | Q#  | ANS | Q#   | ANS |
|-----|-----|-----|-----|-----|-----|-----|-----|------|-----|
| Q1  | B   | Q23 | A   | Q45 | A   | Q67 | B   | Q89  | B   |
| Q2  | D   | Q24 | C   | Q46 | D   | Q68 | C   | Q90  | A   |
| Q3  | A   | Q25 | A   | Q47 | B   | Q69 | B   | Q91  | D   |
| Q4  | C   | Q26 | D   | Q48 | C   | Q70 | A   | Q92  | C   |
| Q5  | A   | Q27 | B   | Q49 | B   | Q71 | D   | Q93  | A   |
| Q6  | D   | Q28 | C   | Q50 | A   | Q72 | C   | Q94  | C   |
| Q7  | B   | Q29 | B   | Q51 | D   | Q73 | A   | Q95  | B   |
| Q8  | C   | Q30 | A   | Q52 | C   | Q74 | C   | Q96  | D   |
| Q9  | B   | Q31 | D   | Q53 | A   | Q75 | B   | Q97  | C   |
| Q10 | A   | Q32 | C   | Q54 | C   | Q76 | D   | Q98  | A   |
| Q11 | D   | Q33 | A   | Q55 | B   | Q77 | C   | Q99  | D   |
| Q12 | C   | Q34 | C   | Q56 | D   | Q78 | A   | Q100 | B   |
| Q13 | A   | Q35 | B   | Q57 | C   | Q79 | D   | Q101 | C   |
| Q14 | C   | Q36 | D   | Q58 | A   | Q80 | B   | Q102 | A   |
| Q15 | B   | Q37 | C   | Q59 | D   | Q81 | B   | Q103 | D   |
| Q16 | D   | Q38 | A   | Q60 | B   | Q82 | D   | Q104 | B   |
| Q17 | C   | Q39 | D   | Q61 | B   | Q83 | A   | Q105 | A   |
| Q18 | A   | Q40 | B   | Q62 | D   | Q84 | C   | Q106 | C   |
| Q19 | D   | Q41 | B   | Q63 | A   | Q85 | A   | Q107 | B   |
| Q20 | B   | Q42 | D   | Q64 | C   | Q86 | D   | Q108 | D   |
| Q21 | B   | Q43 | A   | Q65 | A   | Q87 | B   | Q109 | A   |
| Q22 | D   | Q44 | C   | Q66 | D   | Q88 | C   | Q110 | C   |

## Explanations & Distractor Analysis — Topic 24 — Electrochemistry (A Level MCQs)

CAMBRIDGE INTERNATIONAL A LEVEL CHEMISTRY (9701) — EXAMINER RATIONALE

|    |          |                                                                                                                                                                                                                                                                                                                                                 |     |
|----|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 1  | <b>B</b> | Option B is correct. The Faraday constant (F) represents the electric charge carried by one mole of electrons: $F = L \times e = (6.02 \times 10^{23} \text{ mol}^{-1}) \times (1.60 \times 10^{-19} \text{ C}) \approx 96500 \text{ C mol}^{-1}$ .                                                                                             |     |
| 2  | <b>D</b> | Option D is correct. At cathode: $4\text{H}^+ + 4\text{e}^- \rightarrow 2\text{H}_2$ (2 moles $\text{H}_2$ ). At anode: $2\text{H}_2\text{O} \rightarrow \text{O}_2 + 4\text{H}^+ + 4\text{e}^-$ (1 mole $\text{O}_2$ ). Four moles of electrons produce 2 moles of $\text{H}_2$ gas and 1 mole of $\text{O}_2$ gas, giving a 2:1 volume ratio. |     |
| 3  | <b>A</b> | Option A is correct. $Q = I \times t = 2.00 \times 1930 = 3860 \text{ C}$ . Moles of $\text{e}^- = 3860 / 96500 = 0.0400 \text{ mol}$ . For $\text{Ag}^+ + \text{e}^- \rightarrow \text{Ag}$ , moles of $\text{Ag} = 0.0400 \text{ mol}$ . Mass of $\text{Ag} = 0.0400 \times 108.0 = 4.32 \text{ g}$ .                                         |     |
| 4  | <b>C</b> | Option C is correct. To calculate $L = (Q \times Ar) / (m \times z \times e)$ : you need current and time (for Q), mass gained (m), known Ar of copper, charge on electron e, and $z = 2$ . The solution volume does not enter into Faraday's equation.                                                                                         | [1] |
| 5  | <b>A</b> | Option A is correct. Connected in series means identical charge Q (and moles of electrons) passes through both cells. Since $\text{Ag}^+$ requires 1 electron per atom and $\text{Cu}^{2+}$ requires 2 electrons per atom, $n(\text{Cu}) = 0.5 \times n(\text{Ag}) = 0.50 \text{ mol}$ .                                                        | [1] |
| 6  | <b>D</b> | Option D is correct. Connected in series means identical charge Q (and moles of electrons) passes through both cells. Since $\text{Ag}^+$ requires 1 electron per atom and $\text{Cu}^{2+}$ requires 2 electrons per atom, $n(\text{Cu}) = 0.5 \times n(\text{Ag}) = 0.50 \text{ mol}$ .                                                        | [1] |
| 7  | <b>B</b> | Option B is correct. Connected in series means identical charge Q (and moles of electrons) passes through both cells. Since $\text{Ag}^+$ requires 1 electron per atom and $\text{Cu}^{2+}$ requires 2 electrons per atom, $n(\text{Cu}) = 0.5 \times n(\text{Ag}) = 0.50 \text{ mol}$ .                                                        | [1] |
| 8  | <b>C</b> | Option C is correct. Connected in series means identical charge Q (and moles of electrons) passes through both cells. Since $\text{Ag}^+$ requires 1 electron per atom and $\text{Cu}^{2+}$ requires 2 electrons per atom, $n(\text{Cu}) = 0.5 \times n(\text{Ag}) = 0.50 \text{ mol}$ .                                                        | [1] |
| 9  | <b>B</b> | Option B is correct. Connected in series means identical charge Q (and moles of electrons) passes through both cells. Since $\text{Ag}^+$ requires 1 electron per atom and $\text{Cu}^{2+}$ requires 2 electrons per atom, $n(\text{Cu}) = 0.5 \times n(\text{Ag}) = 0.50 \text{ mol}$ .                                                        | [1] |
| 10 | <b>A</b> | Option A is correct. Connected in series means identical charge Q (and moles of electrons) passes through both cells. Since $\text{Ag}^+$ requires 1 electron per atom and $\text{Cu}^{2+}$ requires 2 electrons per atom, $n(\text{Cu}) = 0.5 \times n(\text{Ag}) = 0.50 \text{ mol}$ .                                                        | [1] |
| 11 | <b>D</b> | Option D is correct. Connected in series means identical charge Q (and moles of electrons) passes through both cells. Since $\text{Ag}^+$ requires 1 electron per atom and $\text{Cu}^{2+}$ requires 2 electrons per atom, $n(\text{Cu}) = 0.5 \times n(\text{Ag}) = 0.50 \text{ mol}$ .                                                        | [1] |
| 12 | <b>C</b> | Option C is correct. Connected in series means identical charge Q (and moles of electrons) passes through both cells. Since $\text{Ag}^+$ requires 1 electron per atom and $\text{Cu}^{2+}$ requires 2 electrons per atom, $n(\text{Cu}) = 0.5 \times n(\text{Ag}) = 0.50 \text{ mol}$ .                                                        | [1] |
| 13 | <b>A</b> | Option A is correct. Connected in series means identical charge Q (and moles of electrons) passes through both cells. Since $\text{Ag}^+$ requires 1 electron per atom and $\text{Cu}^{2+}$ requires 2 electrons per atom, $n(\text{Cu}) = 0.5 \times n(\text{Ag}) = 0.50 \text{ mol}$ .                                                        | [1] |
| 14 | <b>C</b> | Option C is correct. Connected in series means identical charge Q (and moles of electrons) passes through both cells. Since $\text{Ag}^+$ requires 1 electron per atom and $\text{Cu}^{2+}$ requires 2 electrons per atom, $n(\text{Cu}) = 0.5 \times n(\text{Ag}) = 0.50 \text{ mol}$ .                                                        | [1] |
| 15 | <b>B</b> | Option B is correct. Connected in series means identical charge Q (and moles of electrons) passes through both cells. Since $\text{Ag}^+$ requires 1 electron per atom and $\text{Cu}^{2+}$ requires 2 electrons per atom, $n(\text{Cu}) = 0.5 \times n(\text{Ag}) = 0.50 \text{ mol}$ .                                                        | [1] |
| 16 | <b>D</b> | Option D is correct. Connected in series means identical charge Q (and moles of electrons) passes through both cells. Since $\text{Ag}^+$ requires 1 electron per atom and $\text{Cu}^{2+}$ requires 2 electrons per atom, $n(\text{Cu}) = 0.5 \times n(\text{Ag}) = 0.50 \text{ mol}$ .                                                        | [1] |
| 17 | <b>C</b> | Option C is correct. Connected in series means identical charge Q (and moles of electrons) passes through both cells. Since $\text{Ag}^+$ requires 1 electron per atom and $\text{Cu}^{2+}$ requires 2 electrons per atom, $n(\text{Cu}) = 0.5 \times n(\text{Ag}) = 0.50 \text{ mol}$ .                                                        | [1] |
| 18 | <b>A</b> | Option A is correct. Connected in series means identical charge Q (and moles of electrons) passes through both cells. Since $\text{Ag}^+$ requires 1 electron per atom and $\text{Cu}^{2+}$ requires 2 electrons per atom, $n(\text{Cu}) = 0.5 \times n(\text{Ag}) = 0.50 \text{ mol}$ .                                                        | [1] |
| 19 | <b>D</b> | Option D is correct. Connected in series means identical charge Q (and moles of electrons) passes through both cells. Since $\text{Ag}^+$ requires 1 electron per atom and $\text{Cu}^{2+}$ requires 2 electrons per atom, $n(\text{Cu}) = 0.5 \times n(\text{Ag}) = 0.50 \text{ mol}$ .                                                        | [1] |
| 20 | <b>B</b> | Option B is correct. Connected in series means identical charge Q (and moles of electrons) passes through both cells. Since $\text{Ag}^+$ requires 1 electron per atom and $\text{Cu}^{2+}$ requires 2 electrons per atom, $n(\text{Cu}) = 0.5 \times n(\text{Ag}) = 0.50 \text{ mol}$ .                                                        | [1] |
| 21 | <b>B</b> | Option B is correct. Connected in series means identical charge Q (and moles of electrons) passes through both cells. Since $\text{Ag}^+$ requires 1 electron per atom and $\text{Cu}^{2+}$ requires 2 electrons per atom, $n(\text{Cu}) = 0.5 \times n(\text{Ag}) = 0.50 \text{ mol}$ .                                                        | [1] |
| 22 | <b>D</b> | Option D is correct. Connected in series means identical charge Q (and moles of electrons) passes through both cells. Since $\text{Ag}^+$ requires 1 electron per atom and $\text{Cu}^{2+}$ requires 2 electrons per atom, $n(\text{Cu}) = 0.5 \times n(\text{Ag}) = 0.50 \text{ mol}$ .                                                        | [1] |
| 23 | <b>A</b> | Option A is correct. Connected in series means identical charge Q (and moles of electrons) passes through both cells. Since $\text{Ag}^+$ requires 1 electron per atom and $\text{Cu}^{2+}$ requires 2 electrons per atom, $n(\text{Cu}) = 0.5 \times n(\text{Ag}) = 0.50 \text{ mol}$ .                                                        | [1] |
| 24 | <b>C</b> | Option C is correct. Connected in series means identical charge Q (and moles of electrons) passes through both cells. Since $\text{Ag}^+$ requires 1 electron per atom and $\text{Cu}^{2+}$ requires 2 electrons per atom, $n(\text{Cu}) = 0.5 \times n(\text{Ag}) = 0.50 \text{ mol}$ .                                                        | [1] |
| 25 | <b>A</b> | Option A is correct. Connected in series means identical charge Q (and moles of electrons) passes through both cells. Since $\text{Ag}^+$ requires 1 electron per atom and $\text{Cu}^{2+}$ requires 2 electrons per atom, $n(\text{Cu}) = 0.5 \times n(\text{Ag}) = 0.50 \text{ mol}$ .                                                        | [1] |
| 26 | <b>D</b> | Option D is correct. Connected in series means identical charge Q (and moles of electrons) passes through both cells. Since $\text{Ag}^+$ requires 1 electron per atom and $\text{Cu}^{2+}$ requires 2 electrons per atom, $n(\text{Cu}) = 0.5 \times n(\text{Ag}) = 0.50 \text{ mol}$ .                                                        | [1] |

|    |          |                                                                                                                                                                                                                                                                                                                                                                    |     |
|----|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 27 | <b>B</b> | Option B is correct. Connected in series means identical charge $Q$ (and moles of electrons) passes through both cells. Since $\text{Ag}^+$ requires 1 electron per atom and $\text{Cu}^{2+}$ requires 2 electrons per atom, $n(\text{Cu}) = 0.5 \times n(\text{Ag}) = 0.50 \text{ mol}$ .                                                                         | [1] |
| 28 | <b>C</b> | Option C is correct. Connected in series means identical charge $Q$ (and moles of electrons) passes through both cells. Since $\text{Ag}^+$ requires 1 electron per atom and $\text{Cu}^{2+}$ requires 2 electrons per atom, $n(\text{Cu}) = 0.5 \times n(\text{Ag}) = 0.50 \text{ mol}$ .                                                                         | [1] |
| 29 | <b>B</b> | Option B is correct. Connected in series means identical charge $Q$ (and moles of electrons) passes through both cells. Since $\text{Ag}^+$ requires 1 electron per atom and $\text{Cu}^{2+}$ requires 2 electrons per atom, $n(\text{Cu}) = 0.5 \times n(\text{Ag}) = 0.50 \text{ mol}$ .                                                                         | [1] |
| 30 | <b>A</b> | Option A is correct. Connected in series means identical charge $Q$ (and moles of electrons) passes through both cells. Since $\text{Ag}^+$ requires 1 electron per atom and $\text{Cu}^{2+}$ requires 2 electrons per atom, $n(\text{Cu}) = 0.5 \times n(\text{Ag}) = 0.50 \text{ mol}$ .                                                                         | [1] |
| 31 | <b>D</b> | Option D is correct. Connected in series means identical charge $Q$ (and moles of electrons) passes through both cells. Since $\text{Ag}^+$ requires 1 electron per atom and $\text{Cu}^{2+}$ requires 2 electrons per atom, $n(\text{Cu}) = 0.5 \times n(\text{Ag}) = 0.50 \text{ mol}$ .                                                                         | [1] |
| 32 | <b>C</b> | Option C is correct. Connected in series means identical charge $Q$ (and moles of electrons) passes through both cells. Since $\text{Ag}^+$ requires 1 electron per atom and $\text{Cu}^{2+}$ requires 2 electrons per atom, $n(\text{Cu}) = 0.5 \times n(\text{Ag}) = 0.50 \text{ mol}$ .                                                                         | [1] |
| 33 | <b>A</b> | Option A is correct. Connected in series means identical charge $Q$ (and moles of electrons) passes through both cells. Since $\text{Ag}^+$ requires 1 electron per atom and $\text{Cu}^{2+}$ requires 2 electrons per atom, $n(\text{Cu}) = 0.5 \times n(\text{Ag}) = 0.50 \text{ mol}$ .                                                                         | [1] |
| 34 | <b>C</b> | Option C is correct. Connected in series means identical charge $Q$ (and moles of electrons) passes through both cells. Since $\text{Ag}^+$ requires 1 electron per atom and $\text{Cu}^{2+}$ requires 2 electrons per atom, $n(\text{Cu}) = 0.5 \times n(\text{Ag}) = 0.50 \text{ mol}$ .                                                                         | [1] |
| 35 | <b>B</b> | Option B is correct. Connected in series means identical charge $Q$ (and moles of electrons) passes through both cells. Since $\text{Ag}^+$ requires 1 electron per atom and $\text{Cu}^{2+}$ requires 2 electrons per atom, $n(\text{Cu}) = 0.5 \times n(\text{Ag}) = 0.50 \text{ mol}$ .                                                                         | [1] |
| 36 | <b>D</b> | Option D is correct. Connected in series means identical charge $Q$ (and moles of electrons) passes through both cells. Since $\text{Ag}^+$ requires 1 electron per atom and $\text{Cu}^{2+}$ requires 2 electrons per atom, $n(\text{Cu}) = 0.5 \times n(\text{Ag}) = 0.50 \text{ mol}$ .                                                                         | [1] |
| 37 | <b>C</b> | Option C is correct. Connected in series means identical charge $Q$ (and moles of electrons) passes through both cells. Since $\text{Ag}^+$ requires 1 electron per atom and $\text{Cu}^{2+}$ requires 2 electrons per atom, $n(\text{Cu}) = 0.5 \times n(\text{Ag}) = 0.50 \text{ mol}$ .                                                                         | [1] |
| 38 | <b>A</b> | Option A is correct. Connected in series means identical charge $Q$ (and moles of electrons) passes through both cells. Since $\text{Ag}^+$ requires 1 electron per atom and $\text{Cu}^{2+}$ requires 2 electrons per atom, $n(\text{Cu}) = 0.5 \times n(\text{Ag}) = 0.50 \text{ mol}$ .                                                                         | [1] |
| 39 | <b>D</b> | Option D is correct. Connected in series means identical charge $Q$ (and moles of electrons) passes through both cells. Since $\text{Ag}^+$ requires 1 electron per atom and $\text{Cu}^{2+}$ requires 2 electrons per atom, $n(\text{Cu}) = 0.5 \times n(\text{Ag}) = 0.50 \text{ mol}$ .                                                                         | [1] |
| 40 | <b>B</b> | Option B is correct. Connected in series means identical charge $Q$ (and moles of electrons) passes through both cells. Since $\text{Ag}^+$ requires 1 electron per atom and $\text{Cu}^{2+}$ requires 2 electrons per atom, $n(\text{Cu}) = 0.5 \times n(\text{Ag}) = 0.50 \text{ mol}$ .                                                                         | [1] |
| 41 | <b>B</b> | Option B is correct. Connected in series means identical charge $Q$ (and moles of electrons) passes through both cells. Since $\text{Ag}^+$ requires 1 electron per atom and $\text{Cu}^{2+}$ requires 2 electrons per atom, $n(\text{Cu}) = 0.5 \times n(\text{Ag}) = 0.50 \text{ mol}$ .                                                                         | [1] |
| 42 | <b>D</b> | Option D is correct. Connected in series means identical charge $Q$ (and moles of electrons) passes through both cells. Since $\text{Ag}^+$ requires 1 electron per atom and $\text{Cu}^{2+}$ requires 2 electrons per atom, $n(\text{Cu}) = 0.5 \times n(\text{Ag}) = 0.50 \text{ mol}$ .                                                                         | [1] |
| 43 | <b>A</b> | Option A is correct. Connected in series means identical charge $Q$ (and moles of electrons) passes through both cells. Since $\text{Ag}^+$ requires 1 electron per atom and $\text{Cu}^{2+}$ requires 2 electrons per atom, $n(\text{Cu}) = 0.5 \times n(\text{Ag}) = 0.50 \text{ mol}$ .                                                                         | [1] |
| 44 | <b>C</b> | Option C is correct. Connected in series means identical charge $Q$ (and moles of electrons) passes through both cells. Since $\text{Ag}^+$ requires 1 electron per atom and $\text{Cu}^{2+}$ requires 2 electrons per atom, $n(\text{Cu}) = 0.5 \times n(\text{Ag}) = 0.50 \text{ mol}$ .                                                                         | [1] |
| 45 | <b>A</b> | Option A is correct. Connected in series means identical charge $Q$ (and moles of electrons) passes through both cells. Since $\text{Ag}^+$ requires 1 electron per atom and $\text{Cu}^{2+}$ requires 2 electrons per atom, $n(\text{Cu}) = 0.5 \times n(\text{Ag}) = 0.50 \text{ mol}$ .                                                                         | [1] |
| 46 | <b>D</b> | Option D is correct. Connected in series means identical charge $Q$ (and moles of electrons) passes through both cells. Since $\text{Ag}^+$ requires 1 electron per atom and $\text{Cu}^{2+}$ requires 2 electrons per atom, $n(\text{Cu}) = 0.5 \times n(\text{Ag}) = 0.50 \text{ mol}$ .                                                                         | [1] |
| 47 | <b>B</b> | Option B is correct. Connected in series means identical charge $Q$ (and moles of electrons) passes through both cells. Since $\text{Ag}^+$ requires 1 electron per atom and $\text{Cu}^{2+}$ requires 2 electrons per atom, $n(\text{Cu}) = 0.5 \times n(\text{Ag}) = 0.50 \text{ mol}$ .                                                                         | [1] |
| 48 | <b>C</b> | Option C is correct. Connected in series means identical charge $Q$ (and moles of electrons) passes through both cells. Since $\text{Ag}^+$ requires 1 electron per atom and $\text{Cu}^{2+}$ requires 2 electrons per atom, $n(\text{Cu}) = 0.5 \times n(\text{Ag}) = 0.50 \text{ mol}$ .                                                                         | [1] |
| 49 | <b>B</b> | Option B is correct. Connected in series means identical charge $Q$ (and moles of electrons) passes through both cells. Since $\text{Ag}^+$ requires 1 electron per atom and $\text{Cu}^{2+}$ requires 2 electrons per atom, $n(\text{Cu}) = 0.5 \times n(\text{Ag}) = 0.50 \text{ mol}$ .                                                                         | [1] |
| 50 | <b>A</b> | Option A is correct. Connected in series means identical charge $Q$ (and moles of electrons) passes through both cells. Since $\text{Ag}^+$ requires 1 electron per atom and $\text{Cu}^{2+}$ requires 2 electrons per atom, $n(\text{Cu}) = 0.5 \times n(\text{Ag}) = 0.50 \text{ mol}$ .                                                                         | [1] |
| 51 | <b>D</b> | Option D is correct. By international IUPAC convention, the standard hydrogen electrode ( $2\text{H}^+(\text{aq}) + 2\text{e}^- \rightleftharpoons \text{H}_2(\text{g})$ ) at 298 K, 100 kPa $\text{H}_2$ pressure, and $1.0 \text{ mol dm}^{-3} [\text{H}^+]$ is defined as the zero reference point: $E^\circ = 0.00 \text{ V}$ .                                | [1] |
| 52 | <b>C</b> | Option C is correct. $G^\circ = -nFE^\circ_{\text{cell}}$ . Therefore, when $E^\circ_{\text{cell}}$ is positive, $G^\circ$ is negative, which corresponds to thermodynamic feasibility.                                                                                                                                                                            | [1] |
| 53 | <b>A</b> | Option A is correct. By Le Chatelier's principle, increasing $[\text{Zn}^{2+}]$ shifts the equilibrium $\text{Zn}^{2+} + 2\text{e}^- \rightleftharpoons \text{Zn}$ to the right, accepting electrons more readily and making the potential more positive (less negative). Quantitatively: $E = E^\circ + (0.059/2) \log(2.0) = -0.76 + 0.009 = -0.751 \text{ V}$ . | [1] |

MENTORA ACADEMY · Urwah · A Level Multiple Choice Pack – Topic 24 – Electrochemistry (A Level MCQs)

|     |          |                                                                                                                                                                                                                                     |     |
|-----|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 87  | <b>B</b> | Option B is correct. $E_{\text{cell}} = (0.059 / 1) \log_{10}([\text{Ag}^+]_{\text{cathode}} / [\text{Ag}^+]_{\text{anode}}) = 0.059 \times \log_{10}(1.0 / 0.01) = 0.059 \times 2 = +0.118 \text{ V}$ .                            | [1] |
| 88  | <b>C</b> | Option C is correct. $E_{\text{cell}} = (0.059 / 1) \log_{10}([\text{Ag}^+]_{\text{cathode}} / [\text{Ag}^+]_{\text{anode}}) = 0.059 \times \log_{10}(1.0 / 0.01) = 0.059 \times 2 = +0.118 \text{ V}$ .                            | [1] |
| 89  | <b>B</b> | Option B is correct. $E_{\text{cell}} = (0.059 / 1) \log_{10}([\text{Ag}^+]_{\text{cathode}} / [\text{Ag}^+]_{\text{anode}}) = 0.059 \times \log_{10}(1.0 / 0.01) = 0.059 \times 2 = +0.118 \text{ V}$ .                            | [1] |
| 90  | <b>A</b> | Option A is correct. $E_{\text{cell}} = (0.059 / 1) \log_{10}([\text{Ag}^+]_{\text{cathode}} / [\text{Ag}^+]_{\text{anode}}) = 0.059 \times \log_{10}(1.0 / 0.01) = 0.059 \times 2 = +0.118 \text{ V}$ .                            | [1] |
| 91  | <b>D</b> | Option D is correct. $E_{\text{cell}} = (0.059 / 1) \log_{10}([\text{Ag}^+]_{\text{cathode}} / [\text{Ag}^+]_{\text{anode}}) = 0.059 \times \log_{10}(1.0 / 0.01) = 0.059 \times 2 = +0.118 \text{ V}$ .                            | [1] |
| 92  | <b>C</b> | Option C is correct. $E_{\text{cell}} = (0.059 / 1) \log_{10}([\text{Ag}^+]_{\text{cathode}} / [\text{Ag}^+]_{\text{anode}}) = 0.059 \times \log_{10}(1.0 / 0.01) = 0.059 \times 2 = +0.118 \text{ V}$ .                            | [1] |
| 93  | <b>A</b> | Option A is correct. $E_{\text{cell}} = (0.059 / 1) \log_{10}([\text{Ag}^+]_{\text{cathode}} / [\text{Ag}^+]_{\text{anode}}) = 0.059 \times \log_{10}(1.0 / 0.01) = 0.059 \times 2 = +0.118 \text{ V}$ .                            | [1] |
| 94  | <b>C</b> | Option C is correct. $E_{\text{cell}} = (0.059 / 1) \log_{10}([\text{Ag}^+]_{\text{cathode}} / [\text{Ag}^+]_{\text{anode}}) = 0.059 \times \log_{10}(1.0 / 0.01) = 0.059 \times 2 = +0.118 \text{ V}$ .                            | [1] |
| 95  | <b>B</b> | Option B is correct. $E_{\text{cell}} = (0.059 / 1) \log_{10}([\text{Ag}^+]_{\text{cathode}} / [\text{Ag}^+]_{\text{anode}}) = 0.059 \times \log_{10}(1.0 / 0.01) = 0.059 \times 2 = +0.118 \text{ V}$ .                            | [1] |
| 96  | <b>D</b> | Option D is correct. $E_{\text{cell}} = (0.059 / 1) \log_{10}([\text{Ag}^+]_{\text{cathode}} / [\text{Ag}^+]_{\text{anode}}) = 0.059 \times \log_{10}(1.0 / 0.01) = 0.059 \times 2 = +0.118 \text{ V}$ .                            | [1] |
| 97  | <b>C</b> | Option C is correct. $E_{\text{cell}} = (0.059 / 1) \log_{10}([\text{Ag}^+]_{\text{cathode}} / [\text{Ag}^+]_{\text{anode}}) = 0.059 \times \log_{10}(1.0 / 0.01) = 0.059 \times 2 = +0.118 \text{ V}$ .                            | [1] |
| 98  | <b>A</b> | Option A is correct. $E_{\text{cell}} = (0.059 / 1) \log_{10}([\text{Ag}^+]_{\text{cathode}} / [\text{Ag}^+]_{\text{anode}}) = 0.059 \times \log_{10}(1.0 / 0.01) = 0.059 \times 2 = +0.118 \text{ V}$ .                            | [1] |
| 99  | <b>D</b> | Option D is correct. $E_{\text{cell}} = (0.059 / 1) \log_{10}([\text{Ag}^+]_{\text{cathode}} / [\text{Ag}^+]_{\text{anode}}) = 0.059 \times \log_{10}(1.0 / 0.01) = 0.059 \times 2 = +0.118 \text{ V}$ .                            | [1] |
| 100 | <b>B</b> | Option B is correct. $E_{\text{cell}} = (0.059 / 1) \log_{10}([\text{Ag}^+]_{\text{cathode}} / [\text{Ag}^+]_{\text{anode}}) = 0.059 \times \log_{10}(1.0 / 0.01) = 0.059 \times 2 = +0.118 \text{ V}$ .                            | [1] |
| 101 | <b>C</b> | Option C is correct. The charge of one mole of electrons is $F = L \times e$ .                                                                                                                                                      | [1] |
| 102 | <b>A</b> | Option A is correct. High $[\text{Cl}^-]$ shifts the potential and the oxygen overpotential on inert electrodes is substantial, favoring $\text{Cl}_2$ evolution.                                                                   | [1] |
| 103 | <b>D</b> | Option D is correct. Copper deposits quantitatively on the cathode, so measuring the mass increase of the cathode gives the moles of Cu deposited.                                                                                  | [1] |
| 104 | <b>B</b> | Option B is correct. Standard conditions are 298 K, 100 kPa (1 bar), and $1.0 \text{ mol dm}^{-3} [\text{H}^+]$ .                                                                                                                   | [1] |
| 105 | <b>A</b> | Option A is correct. $z$ represents the number of electrons transferred per formula unit in the redox half-reaction.                                                                                                                | [1] |
| 106 | <b>C</b> | Option C is correct. Positive $E^\circ_{\text{cell}}$ implies negative $G^\circ$ and large equilibrium constant $K = \exp(nFE^\circ/RT) > 1$ .                                                                                      | [1] |
| 107 | <b>B</b> | Option B is correct. The salt bridge completes the circuit by permitting counter-ion migration (cations to cathode, anions to anode) to prevent charge buildup.                                                                     | [1] |
| 108 | <b>D</b> | Option D is correct. $E^\circ_{\text{cell}} = E^\circ(\text{reduction}) - E^\circ(\text{oxidation}) = +0.52 - (+0.15) = +0.37 \text{ V}$ .                                                                                          | [1] |
| 109 | <b>A</b> | Option A is correct. Fuel cells bypass the thermal cycle (Carnot efficiency limitation) and directly convert free energy into electrical work.                                                                                      | [1] |
| 110 | <b>C</b> | Option C is correct. Strong $\pi$ -acceptor cyanide ligands have a higher stability constant with $\text{Fe}^{3+}$ than $\text{Fe}^{2+}$ , reducing the chemical activity of $\text{Fe}^{3+}$ and lowering the reduction potential. | [1] |